SG No. 113 D_{2d}^3 $P\bar{4}2_1m$ [tetragonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `2a`, site symmetry: `-4..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,-3,-4]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,-7,-8]`

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2]`
2	$[Y, -X, 0]$	$[0, 0, 0]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-8]`

Table 3: Wyckoff bond: `8c@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[Y, -X, -Z]$	$[0, 0, 0]$	`[3]`
4	$[-Y, X, -Z]$	$[0, 0, 0]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[8]`

* Wyckoff site: `2b`, site symmetry: `-4..`

Table 4: Wyckoff bond: `2a@2b`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,-7,-8]`

Table 5: Wyckoff bond: `4b@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2]`
2	$[Y, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6]`
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8]`

Table 6: Wyckoff bond: `8c@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2c`, site symmetry: `2.mm`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,-7,8]`
2	$[X, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,-4,-5,6]`

Table 8: Wyckoff bond: `2b@2c`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,7,-8]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,-4,5,-6]`

Table 9: Wyckoff bond: `2c@2c`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2,7,8]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,4,5,6]`

Table 10: Wyckoff bond: `4d@2c`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,8]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,7]`
3	$[X, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[3,6]`
4	$[-X, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[4,5]`

Table 11: Wyckoff bond: `4e@2c`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,7]`
2	$[-X, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,8]`

continued ...

Table 11

No.	vector	center	mapping
3	$[-X, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[3,5]
4	$[X, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[4,6]

Table 12: Wyckoff bond: 4f@2c, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	[1,-2]
2	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	[3,-4]
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	[5,-6]
4	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, z\right]$	[7,-8]

Table 13: Wyckoff bond: 8g@2c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[2]
3	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[3]
4	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[4]
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[6]
7	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[7]
8	$[Y, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[8]

* Wyckoff site: 4d, site symmetry: 2..

Table 14: Wyckoff bond: 4a@4d, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	[1,-2]
2	$[Y, -X, 0]$	$[0, 0, -z]$	[3,-4]
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[5,-6]
4	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[7,-8]

Table 15: Wyckoff bond: 4b@4d, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	[1,2]
2	$[0, 0, -Z]$	$[0, 0, -z]$	[3,4]
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[5,6]
4	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[7,8]

Table 16: Wyckoff bond: `8c@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[Y, -X, -Z]$	$[0, 0, -z]$	`[3]`
4	$[-Y, X, -Z]$	$[0, 0, -z]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[7]`
8	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[8]`

* Wyckoff site: `4e`, site symmetry: `..m`

Table 17: Wyckoff bond: `4a@4e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	`[1,8]`
2	$[-X, -X, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[2,7]`
3	$[X, -X, -Z]$	$\left[x+\frac{1}{2}, -x, -z\right]$	`[3,6]`
4	$[-X, X, -Z]$	$\left[\frac{1}{2}-x, x, -z\right]$	`[4,5]`

Table 18: Wyckoff bond: `4b@4e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[x, x+\frac{1}{2}, z\right]$	`[1,-8]`
2	$[-X, X, 0]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[2,-7]`
3	$[-X, -X, 0]$	$\left[x+\frac{1}{2}, -x, -z\right]$	`[3,-6]`
4	$[X, X, 0]$	$\left[\frac{1}{2}-x, x, -z\right]$	`[4,-5]`

Table 19: Wyckoff bond: `8c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[2]`
3	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, -z\right]$	`[3]`
4	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x, -z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -x, -z\right]$	`[6]`
7	$[-Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[7]`
8	$[Y, X, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	`[8]`

* Wyckoff site: `8f`, site symmetry: `1`

Table 20: Wyckoff bond: `8a@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	`[1]`
2	$[-X,\, -Y,\, Z]$	$[-x,\, -y,\, z]$	`[2]`
3	$[Y,\, -X,\, -Z]$	$[y,\, -x,\, -z]$	`[3]`
4	$[-Y,\, X,\, -Z]$	$[-y,\, x,\, -z]$	`[4]`
5	$[-X,\, Y,\, -Z]$	$\left[\frac{1}{2} - x,\, y + \frac{1}{2},\, -z\right]$	`[5]`
6	$[X,\, -Y,\, -Z]$	$\left[x + \frac{1}{2},\, \frac{1}{2} - y,\, -z\right]$	`[6]`
7	$[-Y,\, -X,\, Z]$	$\left[\frac{1}{2} - y,\, \frac{1}{2} - x,\, z\right]$	`[7]`
8	$[Y,\, X,\, Z]$	$\left[y + \frac{1}{2},\, x + \frac{1}{2},\, z\right]$	`[8]`